GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Engineering Mathematics — Official GATE ECE Syllabus

Paper Code: EC • Section 1: Engineering Mathematics (13 Marks Fixed Weightage)

Weightage: 13 Marks • **Question Types:** MCQ, MSQ, NAT • **Difficulty:** Moderate to High

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: LINEAR ALGEBRA

- Vector space, basis, linear dependence and independence of vectors.
- Matrix algebra, algebra of symmetric, skew-symmetric, orthogonal, Hermitian, skew-Hermitian, and unitary matrices.
- Eigenvalues and eigenvectors, algebraic and geometric multiplicity, characteristic equation, Cayley-Hamilton Theorem.
- Rank of a matrix, nullity, rank-nullity theorem, row reduced echelon form (RREF).
- System of linear equations: consistency criteria, existence and uniqueness of solutions (Ax = b and Ax = 0).

SECTION: CALCULUS

- Mean value theorems: Rolle's theorem, Cauchy's mean value theorem, Lagrange's mean value theorem.
- Theorems of integral calculus: Fundamental theorem of calculus, evaluation of definite and improper integrals.
- Partial derivatives, total derivatives, chain rule, Euler's theorem on homogeneous functions.
- Maxima and minima of functions of single and multiple variables, saddle points, method of Lagrange multipliers.
- Multiple integrals: Double and triple integrals, change of order of integration, change of variables (Cartesian to polar/cylindrical/spherical).
- Vector calculus: Gradient, divergence, and curl. Directional derivatives, scalar potential.
- Vector integration theorems: Line, surface, and volume integrals. Green's theorem in the plane, Stokes' theorem, Gauss divergence theorem.

SECTION: DIFFERENTIAL EQUATIONS

- First order equations (linear and non-linear): Integrating factor method, exact differential equations, Bernoulli's equation.
- Higher order linear differential equations with constant coefficients: Complementary functions and particular integrals (shortcuts for exp, sin, cos, polynomials).
- Cauchy-Euler differential equations and transformation to constant coefficient equations.
- Method of variation of parameters for second-order differential equations.
- Initial and boundary value problems in engineering circuits.
- Partial Differential Equations (PDE): Variable separable method, boundary value problems for wave and heat equations.

SECTION: COMPLEX ANALYSIS

- Analytic functions, necessary and sufficient conditions for analyticity: Cauchy-Riemann equations in Cartesian and polar coordinates.
- Harmonic functions and harmonic conjugates, orthogonal trajectories.

- Contour integrals: Cauchy's integral theorem, Cauchy's integral formula for functions and derivatives.
- Taylor's and Laurent's series expansions, classification of singularities (isolated, removable, poles, essential).
- Residue theorem, calculation of residues at simple and multiple poles, evaluation of real definite integrals using contour integration.

SECTION: PROBABILITY AND STATISTICS

- Axiomatic definitions of probability, sampling theorems, conditional probability, independent events, Bayes' Theorem.
- Random variables: Discrete and continuous random variables, Probability Mass Function (PMF), Probability Density Function (PDF), Cumulative Distribution Function (CDF).
- Statistical parameters: Mean, median, mode, variance, standard deviation, moments, and moment generating functions.
- Standard probability distributions: Binomial, Poisson, Uniform, Exponential, and Normal (Gaussian) distributions.
- Joint probability distributions: Joint PDF, marginal distributions, conditional expectation, covariance, correlation coefficient.
- Central limit theorem, Chebyshev's inequality, law of large numbers.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Advanced Engineering Mathematics — Erwin Kreyszig (Wiley)
- Higher Engineering Mathematics — B.S. Grewal (Khanna Publishers)
- Linear Algebra and Its Applications — Gilbert Strang

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- Linear Algebra and Calculus consistently yield 8 out of the 13 marks.
- Eigenvalues of special matrices (Hermitian, Orthogonal, Unitary) are tested almost every year.
- Cauchy's Residue theorem is a guaranteed 2-mark question; master contour poles on and inside unit circles.